Section A.4

We consider two orthonormal bases, $B = \{|e_1\rangle, |e_2\rangle, \ldots, |e_n\rangle\}$ and $B' = \{|f_1\rangle, |f_2\rangle, \ldots, |f_n\rangle\}$

The transformation matrix S is defined by

$|e_j\rangle = \sum_{i=1}^{n} S_{ij} |f_i\rangle$, i.e. the j-th column of S contains the coefficients of $|e_j\rangle$ written out in terms of B'.

Thm: S is unitary

Proof:

$$\delta_{ij} = \langle e_i | e_j \rangle = \langle e_i | \left(\sum_{\beta=1}^{n} S_{\beta j} |f_\beta\rangle \right) = \sum_{\beta=1}^{n} S_{\beta j} \langle f_\beta | e_i \rangle^*$$

$$= \sum_{\beta=1}^{n} S_{\beta j} \left[\langle f_\beta | \left(\sum_{m=1}^{n} S_{mi} |f_m\rangle \right) \right]^* = \sum_{\beta=1}^{n} \sum_{m=1}^{n} S_{\beta j} S^*_{mi} \delta_{\beta m}$$

$$= \sum_{\beta=1}^{n} S_{\beta j} S^*_{\beta i} = \sum_{\beta=1}^{n} S_{\beta j} S^+_{i\beta}$$

$$= (S^+ S)_{ij}$$

QED

Thm: Let S be a linear transformation and $|\alpha\rangle$ and $|\beta\rangle$ be arbitrary vectors.

In (A.48) in the book, the Hermitian conjugate transformation S^+ is defined as that whose matrix elements are $(S^+)_{ij} = S^*_{ji}$.

Then $\langle \alpha | S | \beta \rangle = \langle \beta | S^+ | \alpha \rangle^*$

Proof: We denote the components of $|\alpha\rangle = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix}$ and $|\beta\rangle = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}$

$$\langle \beta | S^+ | \alpha \rangle^* = \left[\begin{pmatrix} b_1^* & \cdots & b_n^* \end{pmatrix} \begin{pmatrix} S_{11}^* & S_{21}^* & \cdots & S_{n1}^* \\ S_{12}^* & & & \\ \vdots & & & \\ S_{1n}^* & \cdots & & S_{nn}^* \end{pmatrix} \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} \right]^* = \left[\begin{pmatrix} b_1^* & \cdots & b_n^* \end{pmatrix} \begin{pmatrix} \sum_{i=1}^{n} S_{i1}^* a_i \\ \vdots \\ \sum_{i=1}^{n} S_{in}^* a_i \end{pmatrix} \right]^*$$

$$= \left[\sum_{j=1}^{n} b_j^* \sum_{i=1}^{n} S_{ij}^* a_i \right]^* = \sum_{i=1}^{n} a_i^* \sum_{j=1}^{m} S_{ij} b_j$$

$$= \begin{pmatrix} a_1^* & \cdots & a_n^* \end{pmatrix} \begin{pmatrix} \sum_{j=1}^{m} S_{1j} b_j \\ \vdots \\ \sum_{j=1}^{m} S_{nj} b_j \end{pmatrix} = \begin{pmatrix} a_1^* & \cdots & a_n^* \end{pmatrix} \begin{pmatrix} S_{11} & S_{12} & \cdots & S_{1m} \\ S_{21} & & & \vdots \\ \vdots & & & \\ S_{n1} & \cdots & & S_{nm} \end{pmatrix} \begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix}$$

$$= \langle \alpha | S | \beta \rangle$$

QED

Example A.1

$$M = \begin{pmatrix} 2 & 0 & -2 \\ -2i & i & 2i \\ 1 & 0 & -1 \end{pmatrix}$$

The characteristic equation is

$$0 = \begin{vmatrix} 2-\lambda & 0 & -2 \\ -2i & i-\lambda & 2i \\ 1 & 0 & -1-\lambda \end{vmatrix} = (i-\lambda)\begin{vmatrix} 2-\lambda & -2 \\ 1 & -1-\lambda \end{vmatrix} = (i-\lambda)\left[(\lambda-2)(\lambda+1)+2\right]$$

$$= (i-\lambda)(\lambda^2-\lambda) = \lambda(i-\lambda)(\lambda-1)$$

The roots are $\lambda = 0$, $\lambda = i$, and $\lambda = 1$

$\lambda = 0$

$$\begin{pmatrix} 2 & 0 & -2 \\ -2i & i & 2i \\ 1 & 0 & -1 \end{pmatrix}\begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \iff \begin{cases} 2a_1 - 2a_3 = 0 \\ -2ia_1 + ia_2 + 2ia_3 = 0 \\ a_1 - a_3 = 0 \end{cases} \implies \begin{cases} a_1 = a_3 \\ a_2 = 0 \end{cases}$$

So eigenvector is $a = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$

$\lambda = 1$

$$\begin{pmatrix} 2 & 0 & -2 \\ -2i & i & 2i \\ 1 & 0 & -1 \end{pmatrix}\begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} \iff \begin{cases} 2a_1 - 2a_3 = a_1 \\ -2ia_1 + ia_2 + 2ia_3 = a_2 \quad (*) \\ a_1 - a_3 = a_3 \end{cases} \implies a_1 = 2a_3$$

$$(*)\ -4ia_3 + 2ia_3 = a_2(1-i) \iff 2ia_3 = a_2(i-1) \iff a_2 = \frac{2i}{i-1}a_3$$

$$\iff a_2 = -i(i+1)a_3 = (1-i)a_3$$

So, the eigenvector is $a = \begin{pmatrix} 2 \\ 1-i \\ 1 \end{pmatrix}$

$\lambda = i$

$$\begin{pmatrix} 2 & 0 & -2 \\ -2i & i & 2i \\ 1 & 0 & -1 \end{pmatrix}\begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = i\begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} \iff \begin{cases} 2a_1 - 2a_3 = ia_1 \\ -2ia_1 + ia_2 + 2ia_3 = ia_2 \\ a_1 - a_3 = ia_3 \end{cases}$$

$$\implies \begin{cases} 2ia_3 = ia_1 \\ -2i(ia_3) + ia_2 = ia_2 \end{cases} \implies a_1 = a_3 = 0 \implies a = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

Derivation on p 479 – p 481:

Thm: Let $T, V \in M^{n\times n}$, both diagonisable. Then,

$$[T,V]=0 \iff \exists\, B=\{v_1,\dots,v_n\} \text{ basis such that } Tv_i=\lambda_i v_i \text{ and } Vv_i=\mu_i v_i$$

Proof:

$\Leftarrow$ If $\mathrm{Spec}(T)$ is non-degenerate:

$$T v_i = \lambda_i v_i$$

$$[T,V]v_i = 0 \iff TVv_i = VTv_i = \lambda_i V v_i$$

$\Rightarrow V v_i$ is an eigenvector of T with the same eigenvalue. But because $\mathrm{Spec}(T)$ is non-degenerate, $Vv_i = k_i v_i$

$\Rightarrow v_i$ is an eigenvector of V.

If $\mathrm{Spec}(T)$ has two-fold degeneracy:

$\exists\, \lambda_0 \in \mathrm{Spec}(T)$, such that v_0 and v_1 are linearly independent, with

$$T v_1 = \lambda_0 v_1 \quad \text{and} \quad T v_2 = \lambda_0 v_2$$

We have $[T,V]v_1 = 0 = [T,V]v_2$. Therefore

$TVv_1 = VTv_1 = \lambda_0 V v_1 \Rightarrow Vv_1$ and Vv_2 are eigenvectors of T with eigenvalue λ_0. It follows that:

$$\begin{cases} Vv_1 = c_{11}v_1 + c_{21}v_2 \\ Vv_2 = c_{12}v_1 + c_{22}v_2 \end{cases}$$

Unless the crossterms vanish, v_1 and v_2 are not eigenvectors of V. But can we find linear combinations of v_1 and v_2 that are eigenvectors of V? If they exist, they would be of the form

$$\begin{cases} w_1 = d_{11}v_1 + d_{21}v_2 \\ w_2 = d_{12}v_1 + d_{22}v_2 \end{cases} \quad \text{with} \quad \begin{cases} Vw_1 = \mu_1 w_1 \\ Vw_2 = \mu_2 w_2 \end{cases}$$

As w_1 and w_2 are linear combinations of v_1 and v_2, they are eigenvectors of T with eigenvalue λ_0. To remain a basis, we also require w_1 and w_2 to be linearly independent.

The claim in the book (equation A.90) is that finding such w_1 and w_2 amounts to solving

$$\begin{pmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{pmatrix} \begin{pmatrix} d_{1i} \\ d_{2i} \end{pmatrix} = \mu_i \begin{pmatrix} d_{1i} \\ d_{2i} \end{pmatrix} \quad (*)$$

which I do in problem A.24. This proves the theorem. But how do we know that (*) has two linearly independent solutions? I.e., how do we know C is diagonisable! This follows from the following lemma:

Thm: Let T be a diagonisable, linear operator from X to X, with $\dim(X) =$ And let $W \subset X$ be a subspace such that $T(w) \in W$ for any $w \in W$. Then $T|_W$ is also diagonisable.

proof: Let $\lambda_1, \dots, \lambda_k$ be the distinct eigenvalues of T, the distinct eigenvalues. I denote the corresponding eigenspaces

$$X_i = \{v \mid Tv = \lambda_i v_i\} = \text{Ker}(T - \lambda_i I)$$

Because T is diagonisable, any $w \in X$ can be written as

$$w = u_1 + \cdots + u_k$$

Step 1. Decompose an arbitrary vector into eigenspace components

Because V is diagonalisable, E has a basis consisting of eigenvectors of V.

Let $\lambda_1, \dots, \lambda_k$ be the **distinct eigenvalues** of V.

Each eigenvalue may have any multiplicity. In particular, degeneracy is allowed.

Denote the corresponding eigenspaces by

$$E_i = \ker(V - \lambda_i I).$$

Since V is diagonalisable, every vector $w \in E$ can be written uniquely as

$$\boxed{w = u_1 + u_2 + \cdots + u_k}$$

where

$$u_i \in E_i, \quad \downarrow \ulcorner u_i = \lambda_i u_i.$$

Below is the theorem and its proof, discussed on p479 – p481.

Thm: Let E be an n-dimensional vector space, and let A and B be be diagonalizable, linear operators. Then,

$$[A,B]=0 \iff \exists\ \mathcal{B}=\{v_1,\dots,v_n\} \text{ a basis for } E \text{ of common eigenvectors of } A \text{ and } B.$$

Proof:

$\boxed{\Leftarrow}$ Let $Av_i=\lambda_i v_i$ and $Bv_i=\mu_i v_i$.

In the basis $\mathcal{B}$, A and B are diagonal. Therefore,

$$AB=\begin{pmatrix}\lambda_1 & & 0\\ & \lambda_2 & \\ 0 & & \ddots \\ & & & \lambda_n\end{pmatrix}\begin{pmatrix}\mu_1 & & 0\\ & \mu_2 & \\ 0 & & \ddots \\ & & & \mu_n\end{pmatrix}=\begin{pmatrix}\lambda_1\mu_1 & & 0\\ & \lambda_2\mu_2 & \\ 0 & & \ddots \\ & & & \lambda_n\mu_n\end{pmatrix}=$$

$$=\begin{pmatrix}\mu_1 & & 0\\ & \mu_2 & \\ 0 & & \ddots \\ & & & \mu_n\end{pmatrix}\begin{pmatrix}\lambda_1 & & 0\\ & \lambda_2 & \\ 0 & & \ddots \\ & & & \lambda_n\end{pmatrix}=BA$$

So A and B commute in this basis. And we show in problem A.24, that if matrices commute in one basis, they commute in any basis.

$\boxed{\Rightarrow}$ For any eigenvalue v of A with eigenvalue λ, i.e. $Av=\lambda v$, we have

$$A(B(v)) \underset{[A,B]=0}{=} B(A(v))=\lambda Bv \qquad (*)$$

and therefore $B(v)$ is either 0, or an eigenvector of A with the same eigenvalue λ.

Let $\lambda_1,\dots,\lambda_k$ be the distinct eigenvalues of A (so that $k\leq n$).

We consider the corresponding eigenspace $E_{\lambda_i}(A)$ for each λ_i, defined as

$$E_{\lambda_i}(A):=\{v\in E \mid Av=\lambda_i v\}$$

Let $\mu_1,\dots,\mu_m$ be the distinct eigenvalues of B.

We now fix an arbitrary $v\in E$. Because A is diagonalizable, we can write

$$v=\sum_{i=1}^{k} a_i \quad \text{with } a_i\in E_{\lambda_i}(A)$$

For each $i\in\{1,\dots,k\}$, we would like to find eigenvectors of B

that lie in $E_{\lambda_i}(A)$ and span it. Is this possible? Take i fixed for now. And let $a \in E_{\lambda_i}(A)$. Because B is diagonalizable, we can write

$$a = \sum_{j=1}^{m} b_j \quad \text{with } b_j \in E_{\mu_j}(B) \qquad (**)$$

We define the polynomial $P_\ell(X) = \prod_{\substack{j=1 \\ j\neq \ell}}^{m} \frac{X-\mu_j}{\mu_\ell - \mu_j}$

Plugging B into this, we get the operator $P_\ell(B) = \prod_{\substack{j=1 \\ j\neq \ell}}^{m} \frac{B-\mu_j I}{\mu_\ell - \mu_j}$.

We then have

$P_\ell(B)(b_j) = \delta_{\ell,j}\, b_\ell$ and therefore

$P_\ell(B)(a) = b_\ell$ for any $\ell \in \{1, 2, \dots, m\}$

We showed in (*) that B maps elements of $E_\lambda(A)$ onto elements of $E_\lambda(A)$. And as $P_\ell(B)$ consists of sums of powers of B and the identity, this is also true for $P_\ell(B)$. This means that because $a \in E_{\lambda_i}(A)$, so is $P_\ell(B)(a) = b_\ell$, for any ℓ. And this in turn means that any (non-zero) element of E_{λ_i} can be written as the sum of simultaneous eigenvectors of A and B (i.e., the sum in (**), chucking out all the zero terms if there are any). Going back to arbitrary $v \in E$, we chose, we can now write it as

$$v = \sum_{i=1}^{k} a_i = \sum_{i=1}^{k} \sum_{j=1}^{m} (b_j)_i \qquad \text{with } (b_j)_i \in E_{\lambda_i}(A) \cap E_{\mu_j}(B).$$

The set $\{(b_j)_i \mid i = 1 \dots k, \text{ and } j = 1 \dots, m\}$ spans E. Removing all the zero elements in it (if there are any), leaves us with a spanning set of common eigenvectors of A and B. As any spanning set can be reduced to a basis, we can thus construct a basis of common eigenvectors of A and B.

QED